

out 24/7 domestic aerial surveillance and help enhance security manifold.

Developed by security major Raytheon, the Joint Land Attack Cruise Missile Defense Elevated Netted Sensor System (JLENS) comprises two 243-foot long aerostats flying 10,000 feet above the ground. One of them has 360-degree surveillance capabilities that enable it to see as far away as 547 km in any direction, while the other has a powerful integrated fire-control radar system. The two aerostats are connected to mobile mooring stations on the ground using tethers that have high-speed fibre-optic cables for sending data as well as transmitting power. The aerostats can also communicate via radio links.

Of course, this announcement has triggered a lot of uproar about privacy concerns, but the government is confident that the two aerostats would work hand in hand to effectively spot and ward off threats.

Human-like, real-time facial recognition

A tool that excels in facial recognition and biometric identification is perhaps what we need to replace the security guard, but how efficient are today's facial recognition tools. It is said that Facebook's facial recognition has near human efficiency, but that is for static images. For real security systems, we need real-time facial recognition. Many companies and research firms are working on such technologies today, and one such example is FST Biometrics (erstwhile FST21), founded by Major General Aharon Zeevi Farkash, the former head of the Israeli Military Intelligence Directorate.

One of their base technologies for In Motion Identification (IMID) is SafeRise, which uses a combination of real-time facial recognition and biometric technology. When somebody walks up to a secured area, the system is capable of immediately recognising whether she is an authorised person or an intruder, without even requiring the person to stop near a camera or provide fingerprints, iris prints, or any authentication.

Mostly, the system relies on facial



Apart from unique facial features, SafeRise also notes your gait, voice and other biometrics (Courtesy: FST Biometrics)

recognition. Using a profile picture, it tries to understand a person's unique features, such as say, the number of pixels between her eyes. Then, every time that person approaches a protected area, the SafeRise system tries to compare all the special features it has stored. If you watch the output of the security camera, the software puts a yellow box over anything that registers as a face. If it recognises the face, the box turns green, else it turns red and the door remains locked. As the system 'watches' the person more, it assimilates more biometric details about her including her gait and voice. The more it learns about the person, the faster it identifies the person. SafeRise is incorporated into FST products like the Digital Doorman.

Quantum leap in mobile phone security

Considering the amount of transactions people are conducting using mobile devices, researchers have now started considering mobile security on the same lines as banking security! A team at the University of Bristol, in collaboration with Nokia, recently demonstrated how quantum cryptography using Quantum Key Distribution (QKD), a bulky and expensive technique currently used for securing fixed physical locations like server rooms in banks, could be used to implement an ultra-high security scheme for mobile devices. New hope comes

from the team's efforts to reduce the bulky and expensive resources required for quantum cryptography, so that it can be implemented by simply integrating an optical chip into a mobile handset.

This is basically a client-server QKD scheme, wherein the cumbersome equipment, such as the laser and detectors, are located at the server side and are accessible to a client over a communication network. The client only requires an on-chip polarisation rotator, which may be integrated into a handheld device. While the experiments seem successful, the team is now working towards implementing the client-server QKD system in a real communications network.

More examples, all around us

There are lot more interesting developments in security across the world—and most of them accurately reflect the changing mindsets of people and the newer threats of networking, increased mobile usage, cloud computing and the Internet of things. For example, a decade ago people would not have thought of do-it-yourself customised home security using sensor-software kits but today people need it to protect their privacy and changing lifestyles. Likewise, nobody would have even imagined that it would become necessary to protect trivial devices like, say, the printer. But, with almost all devices getting connected to the corporate or home network, every device becomes a point of vulnerability.

The threats faced by people today are much more—thanks to the technologies that seemingly promise to make life easier for them, and that means more sleepless nights for technologists paving the way for security X.0! ●

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